

Technical Datasheet 4G Gateway with 4DI, 4DO, & 4AI

Product Overview: -

The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with an integrated I/O interface featuring **4 Digital Inputs (DI)**, **4 Digital Outputs (DO)**, and **4 Analog Inputs (AI)**, alongside a Modbus RS485 port and micro SIM compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



Specifications

General:-

- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access. If 4G is not available, the system automatically switches to Wi-Fi. If both 4G and Wi-Fi are unavailable, the data is securely stored in the SD card for later upload.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **GPS Location Data:** Provides GPS location data for monitoring of remote equipment.
- **Micro Sim Compatibility:** Supports micro-Sim card for flexible and convenient connectivity option.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Dimensions:** 110 mm L x 110 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.

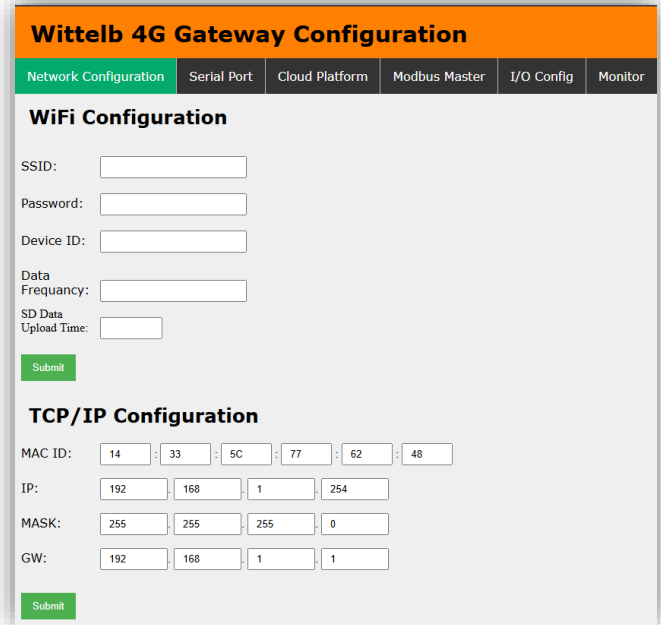
Configuration Settings: -



- Power up the module by connecting a 12-24 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 31) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: Wittelb-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage will be accessible.

Page1:- TCP/IP Configuration:-

- Enter MAC ID of the Ethernet interface (6 hex bytes, e.g. 14:33:5C:77:62:48).
- Enter IP address for the Ethernet port (default: 192.168.1.254).
- Enter Subnet Mask (default: 255.255.255.0).
- Enter Gateway address (default: 192.168.1.1).
- Click Submit to save settings; the module applies to the new Ethernet configuration immediately.



Wittelb 4G Gateway Configuration

Network Configuration | **Serial Port** | Cloud Platform | Modbus Master | I/O Config | Monitor

WiFi Configuration

SSID:

Password:

Device ID:

Data Frequency:

SD Data Upload Time:

TCP/IP Configuration

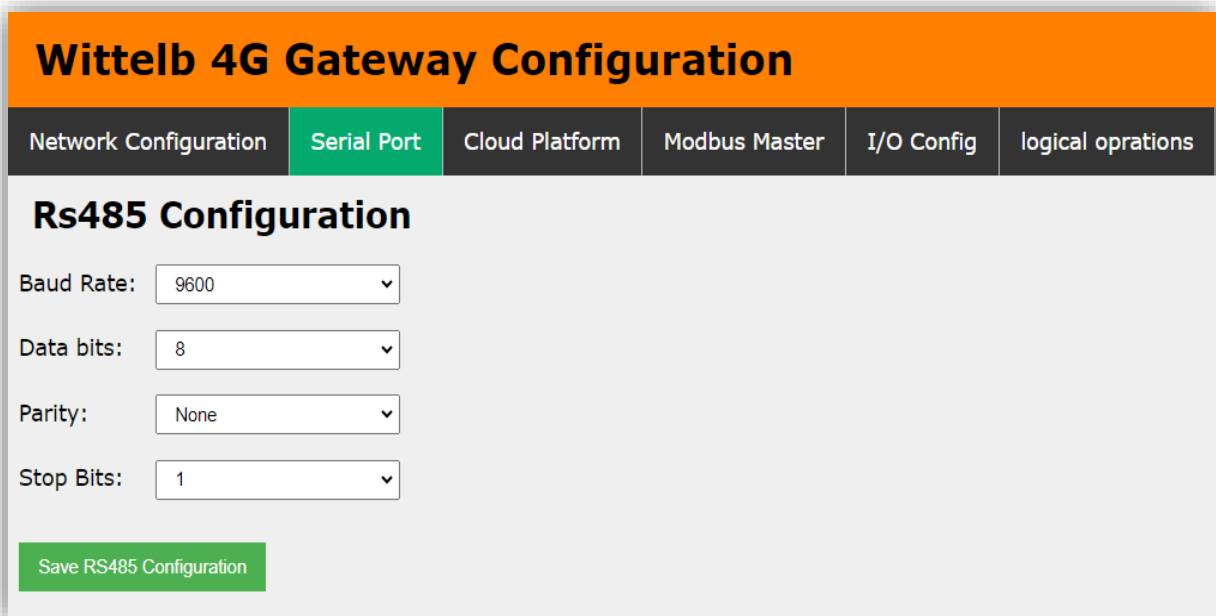
MAC ID: : : : : :

IP: . . .

MASK: . . .

GW: . . .

Page2:- On second page you can set device Baud Rate (9600 / 19200 / 38400 / 57600 / 115200), Data Bits (5 / 6 / 7 / 8), Parity (None / Odd / Even), and Stop Bits (1 / 2) for RS485 Communication



Wittelb 4G Gateway Configuration

Network Configuration | **Serial Port** | Cloud Platform | Modbus Master | I/O Config | logical operations

Rs485 Configuration

Baud Rate:

Data bits:

Parity:

Stop Bits:

Page3:-

- Navigate to page 3 on the cloud platform.
- Input the Host IP or Domain.

Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | **Cloud Platform** | Modbus Master | Monitor | About

Cloud Platform Configuration

Network Mode

WiFi ☒ ☐
 Current Mode: 4G Network

Host IP or Domain:

Port:

Publish Topic:

Publish Topic (SD Card):

Subscribe Topic:

User Name:

Password:

Save Configuration

- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.

Page4:-

Wittelb 4G Gateway Configuration

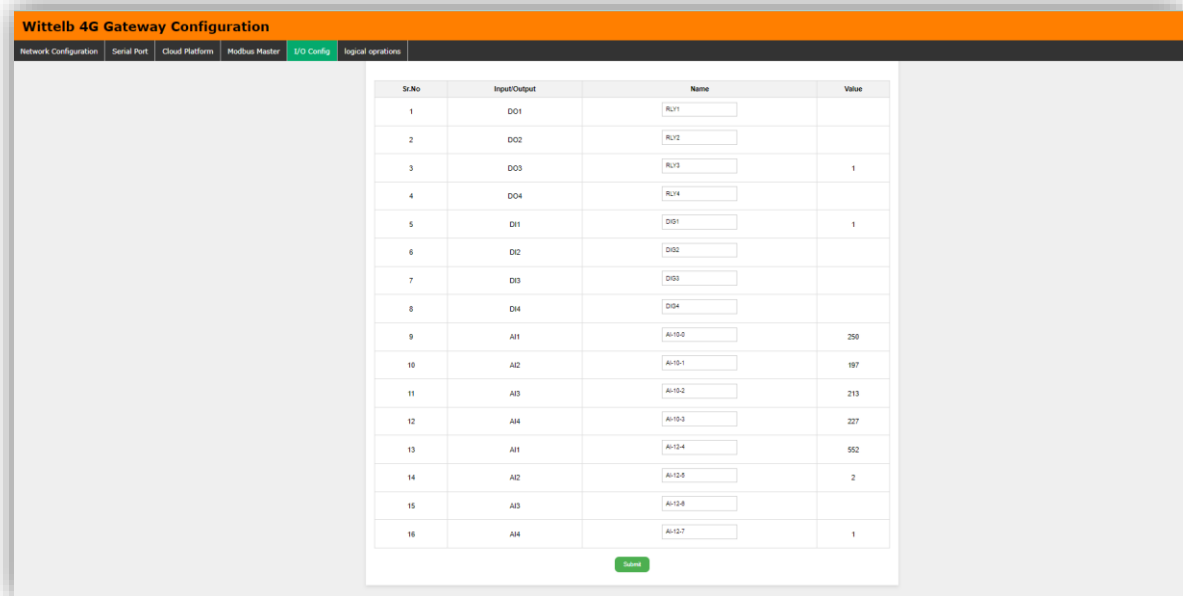
Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | I/O Config | logical operations | About

Name	Slave Address	Register start	Quantity	Function Code	Actions
MFM	5	0	50	3	Delete

Add Row **Submit**

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- Indicate the register end address for the slave.
- Enter the function code corresponding to the desired slave.

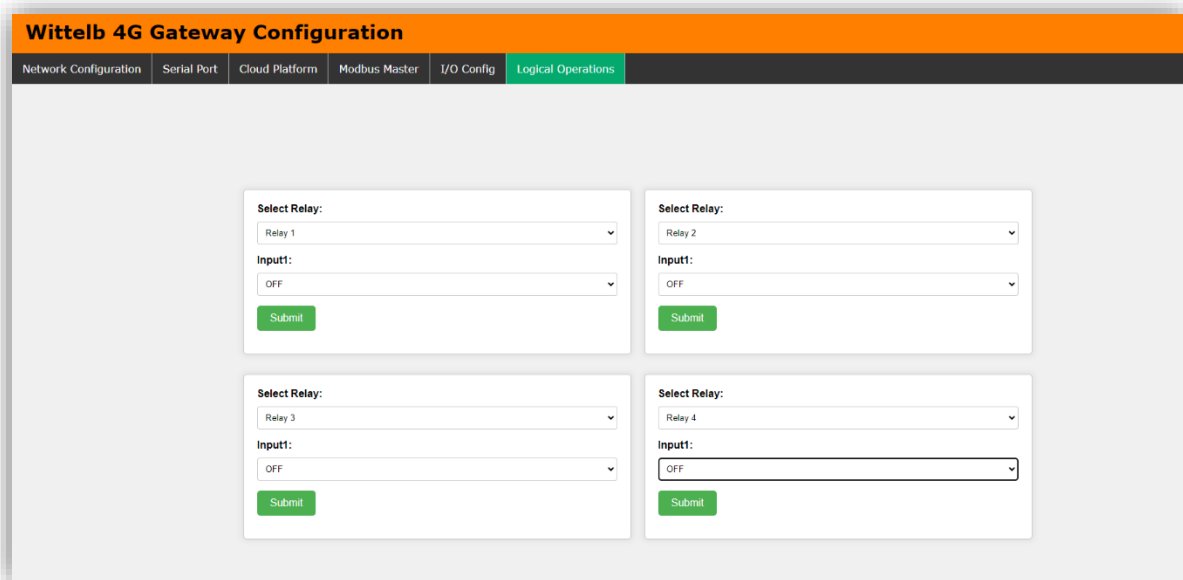
Page5:-



Sl.No	Input/Output	Name	Value
1	DO1	RLY1	
2	DO2	RLY2	
3	DO3	RLY3	1
4	DO4	RLY4	
5	DI1	DI01	1
6	DI2	DI02	
7	DI3	DI03	
8	DI4	DI04	
9	AI1	AI10-0	250
10	AI2	AI10-1	187
11	AI3	AI10-2	213
12	AI4	AI10-3	227
13	AI1	AI12-4	552
14	AI2	AI12-5	2
15	AI3	AI12-6	
16	AI4	AI12-7	1

- Go to page 5 on the I/O Config interface.
- You can view real-time values of both analog and digital inputs on the baseboard.
- Additionally, you have the option to assign specific names to each input/output (I/O) as needed.

Page6:-



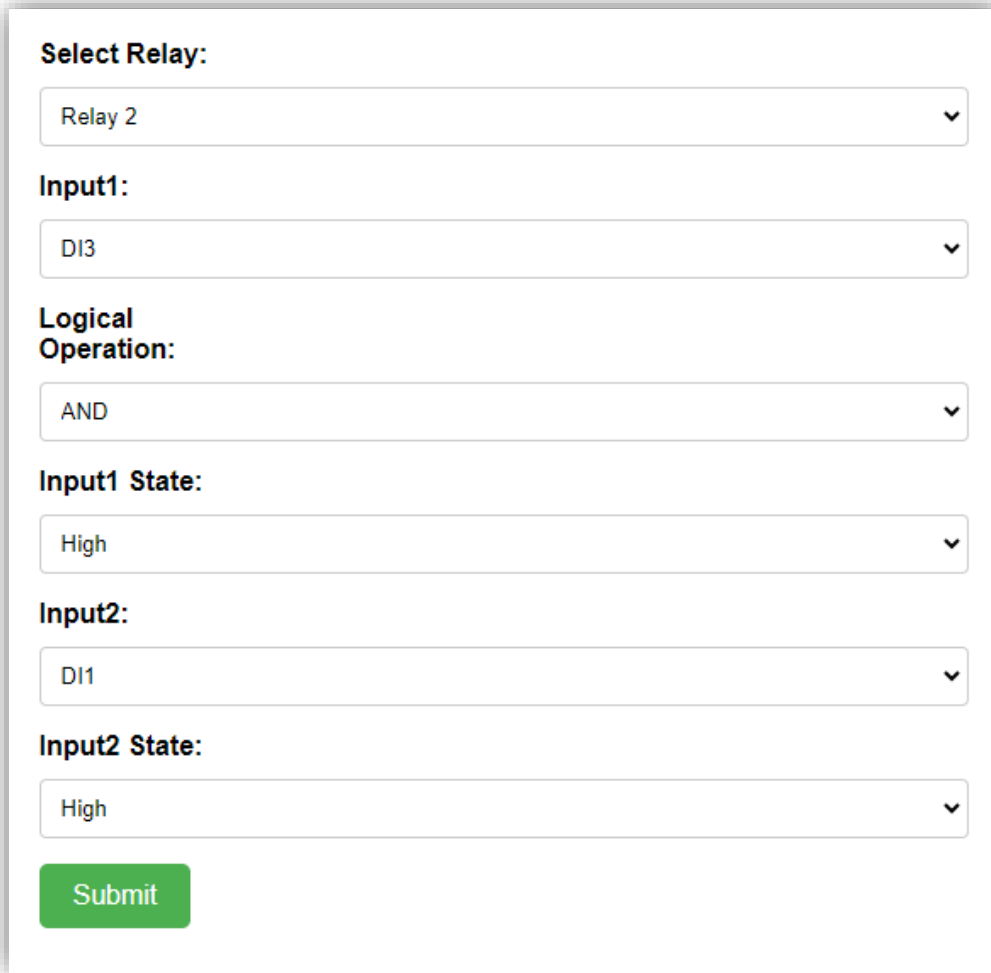
Select Relay:
 Relay 1
Input1:
 OFF
 Submit

Select Relay:
 Relay 2
Input1:
 OFF
 Submit

Select Relay:
 Relay 3
Input1:
 OFF
 Submit

Select Relay:
 Relay 4
Input1:
 OFF
 Submit

- Navigate to page 6 on the Logical Operations interface.



The screenshot displays a configuration window for a relay. It contains several dropdown menus and a submit button. The 'Select Relay:' dropdown is set to 'Relay 2'. The 'Input1:' dropdown is set to 'DI3'. The 'Logical Operation:' dropdown is set to 'AND'. The 'Input1 State:' dropdown is set to 'High'. The 'Input2:' dropdown is set to 'DI1'. The 'Input2 State:' dropdown is set to 'High'. A green 'Submit' button is located at the bottom left of the form.

- Here, you can configure logical AND and OR operations for each of the 4 relays.
- You have the flexibility to assign any two digital inputs to a relay and choose the desired logical operation.
- Additionally, you can assign an analog input to a relay and define minimum and maximum threshold values for relay operation.
- If you prefer not to use any logical operation, simply select the "OFF" option for input one.

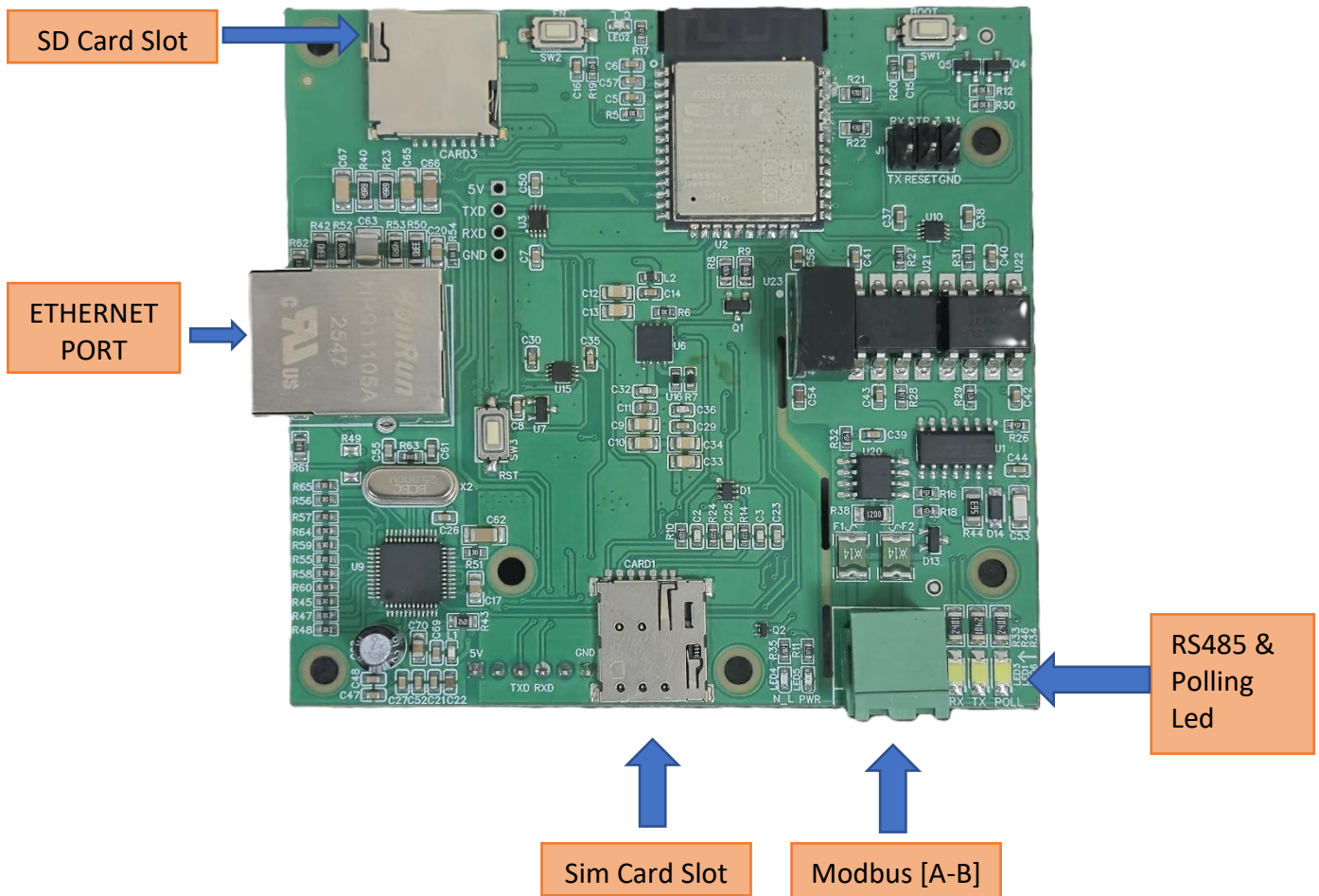
Input & Output module Configuration Settings: -

- The Gateway has 4 Digital Outputs (DO) with Relay of 10 Amps 230V,
- 4 Analog Inputs (AI) to connect with 4-20ma or 0-10 v sensors,

- 4 Digital Inputs (DI) will help monitor the status with logic voltage input between 4-40V DC. Additionally, we have also provided 12 V voltage ref and GND pins for DI detection logic.
- Digital outputs are normally open.
- Data of Digital Inputs (DI) and Analog Inputs (AI) would be directly collected by the Processor and transmitted over 4G to the Cloud based on the Poling frequency.
- Configure each Analog Input (AI) to operate with either 5V or 10V and choose between 10-bit and 12-bit resolution by adjusting the jumper settings.
- You can trigger DO from cloud by sending Following Message

Relay No	ON	OFF
D01	"D01":1	"D01":1
D02	"D02":1	"D01":1
D03	"D03":1	"D01":1
D04	"D04":1	"D01":1

4G Gateway connection Diagram:



Additional Features: -

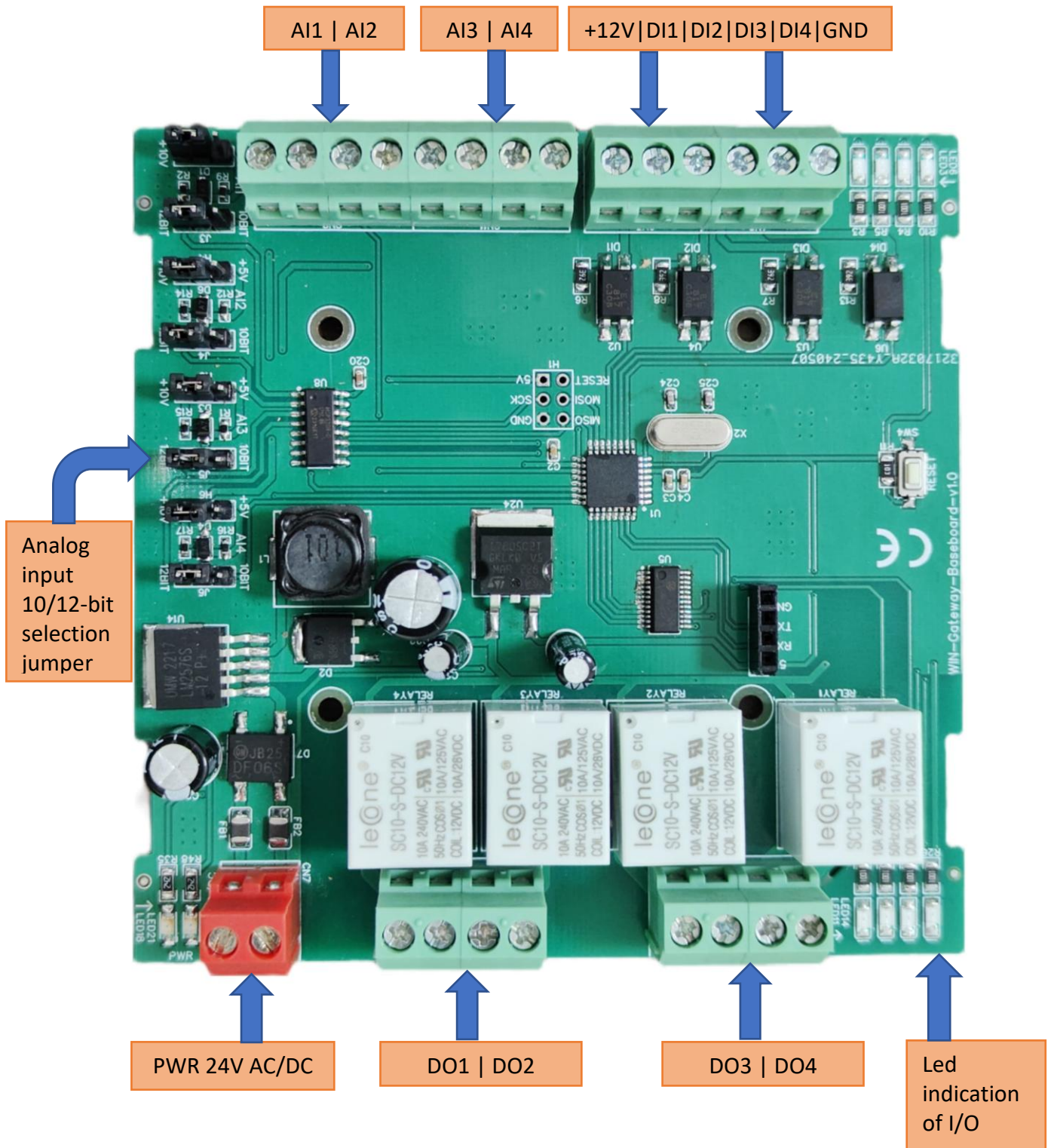
Communication ports are isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

400V isolation.



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